EXPERIMENT NO. 7
CURRENT SENSITIVITY

Aim: To determine the current sensitivity of a moving coil galvanometer.

Apparatus: Moving coil table galvanometer, cell, plug key, reversing key, high resistance box, connecting wires etc.

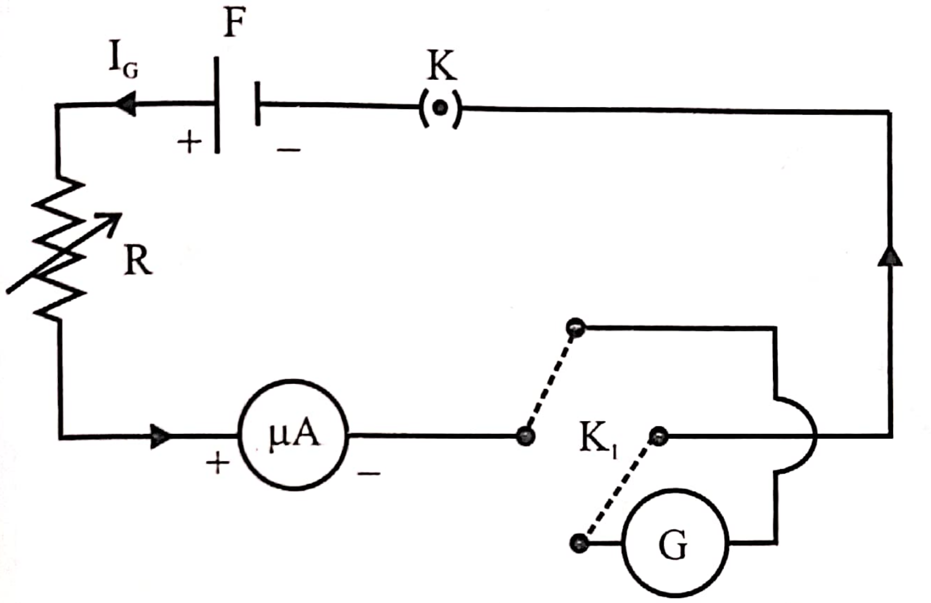
Formula: (i) $S = \dfrac{\theta}{I_G}$

I_G = Current through galvanometer

S = Current sensitivity

θ = deflection in galvanometer

Circuit diagram:



E : Daniel Cell
R : Resistance box
K : key
K_1 : four way key
G : Galvanometer

Procedure:

1. Connect the circuit as shown in figure.
2. Select high resistance from box and close key K and key K_1 so that current flows through galvanometer. Note the deflection θ in the galvanometer.
3. Repeat the procedure for 3 different values of ` R ' so that successive readings differ about by 3 to 4 divisions and it covers entire scale of galvanometer. Note down current I_G in each case.
4. Calculate current sensitivity S for each observation. Find its mean value.
5. Plot a graph of θ against I_G. Hence determine the current sensitivity of the galvanometer

Observations:

(1) Resistance of galvanometer (G) = ..100.. Ω .

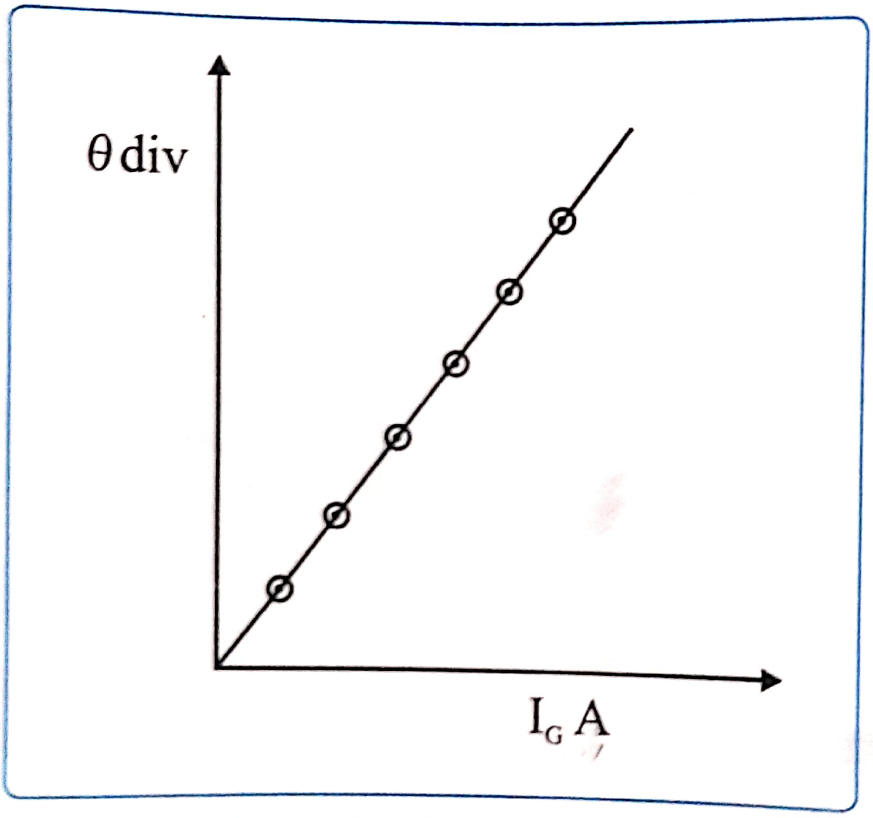
(2) E.M.F. of cell (E) = ..5....volt.

Observation table:

Obs. no	Resistance R Ω	Current I_G µA	Current I_G Ampere	Deflection θ division	Current Sensitivity $S=\frac{\theta}{I_G}$ div / A	Mean S div / A
1.	17,000	100 µA	100×10^{-6} A	6	0.06×10^{6} div/A	
2.	19,000	93 µA	93×10^{-6} A	5	0.053×10^{6} div/A	0.0485x[illegible] div/A
3.	20,000	87 µA	87×10^{-6} A	4	0.0459×10^{6} div/A	
4.	21,000	82 µA	82×10^{-6} A	3	0.0365×10^{6} div/A	

Claculations: Current sensitivity of galvanometer $S=\frac{\theta}{I_G}$ = 0.0485×10^{6} div/A

Graph :



Scale X = 10^{-6} A
Y = θ div

Results : Current sensitivity of a moving coil table galvanometer, (S)

S = 0.0485×10^{6} div/Amp. (by calculation)

S = 0.1666×10^{6} div/Amp. (by graph)

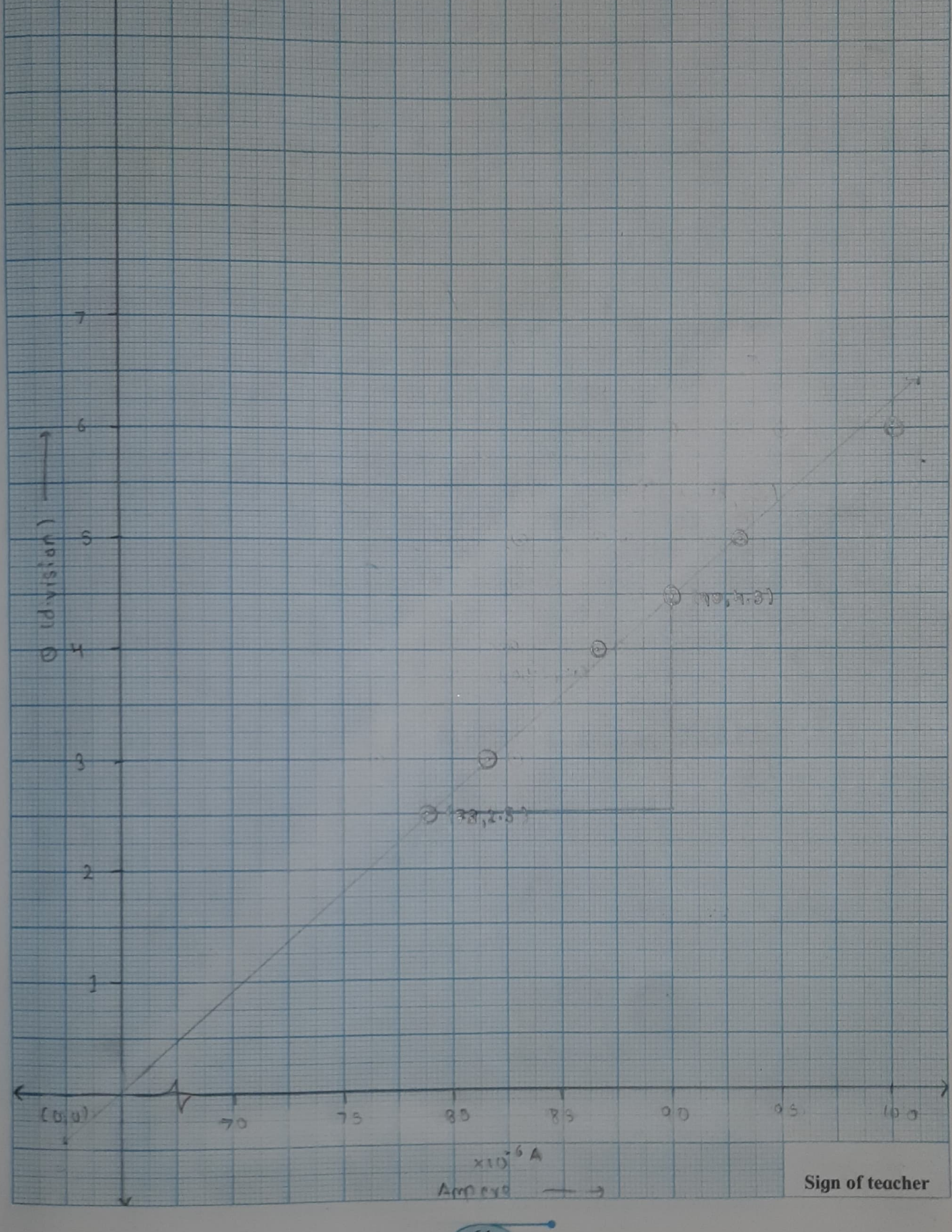
Precautions:

1. Check that all connection at terminal screws are tight.
2. Check that key (K) and key (K_1) are tight.
3. Check that all keys in resistance box are tight except which you remove from the resistance box
4. Check that deflection in the galvanometer when you remove resistance from the resistance box
5. Pass the current through circuit only at the time of observations.
6. Always adjust the resistance in the box so that the pointer galvanometer does not go out of the scale.

Scale :

X-axis :- 2unit = 5×10^{-6} A

Y-axis :- 2unit = 1 θ deflection



Sign of teacher

Questions

1. Define current sensitivity of the galvanometer.

Current sensitivity of a galvanometer is defined as the angular deflection produced per unit current flowing through it.

$$S_I = \frac{\theta}{I} = \frac{NBA}{k}$$

where N is the number of turns, B is the magnetic field, A is the area of the coil, and k is the torsional constant.

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2. State the factors on which current sensitivity depends.

The current sensitivity of a moving coil galvanometer depends on:
Number of turns of the coil N
Magnetic field strength B
Area of the coil A
Torsional constant of the suspension spring k

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3. How can the current sensitivity increase?

The current sensitivity can be increased by:

Increasing the number of turns of the coil
Increasing the area of the coil
Increasing the strength of the magnetic field
Decreasing the torsional constant of the suspension spring (restoring couple)

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Remark and sign of teacher :..
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